

Education

Indian Institute of Technology Patna (IIT Patna)	2023 – 2027
B.Tech. in Mechanical Engineering	4th Year

Technical Skills

Programming:	Python, C++, TypeScript, JavaScript, SQL, HTML, CSS
Web Development:	React, Next.js, FastAPI, Flask, Streamlit, SQLAlchemy, REST APIs
Machine Learning:	Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Cross-Validation, scikit-learn, XGBoost, LightGBM, SHAP, PyTorch, pandas, NumPy
AI & Computer Vision:	Deep Learning, CNN, RNN/LSTM, Transformers, YOLOv11, Object Detection, Multi-Object Tracking, Generative AI, LLMs
NLP & Retrieval:	RAG, Embeddings, Semantic Search, Vector Search, FAISS, Cosine Similarity, Structured Outputs
Databases & Cloud:	PostgreSQL, SQLite, Redis, pgvector, Vector Databases, Docker, Render
Tools & Engineering:	Git, GitHub Actions, MLflow, JWT, WebSockets, Razorpay, Recharts, Zustand, PyMuPDF, pytest

Experience

GatiSetu – AI Mobility Intelligence Platform	May 2025 – Jul 2025
AI Intern	Street Surge Technology GitHub Live

- AI mobility platform for transportation-mode classification and traffic-aware ETA prediction from GPS trajectories; trained a Random Forest (100 trees, depth 12) on 9 kinematic features, achieving **90.9%** hold-out accuracy on 329 labeled segments across 8 modes with macro F1 $\approx$ 0.92.
- Engineered OD matrices and trip segments from cleaned **geospatial GPS data**, using speed, stop ratio, turn rate, and acceleration to generate class probabilities for each trajectory.
- Integrated live traffic with GPS speed for ETA using Google `duration_in_traffic`, OSRM fallback, and IST rush-hour multipliers for 08–11 and 17–21.

Presence	2026
AI / ML Hackathon Project	Accenture Hackathon GitHub Live

- Camera-free occupancy intelligence platform using WiFi multipath to estimate room, people count, and nearby objects; engineered 55 RSSI features over 4-AP, 4 s / 5 Hz windows and trained on 3,472 synthetic samples across 5 rooms and 8 item classes.
- Developed a multi-task supervised learning pipeline using Random Forest (180 trees), HistGradientBoosting, and multi-output RF, achieving **79.0%** exact people-count accuracy (MAE **0.32**) and **78.8%** item Hamming accuracy.
- Applied Kalman filtering and signal processing to reduce RSSI noise and classified sitting/walking/active_group states for camera-free occupancy monitoring and away-mode intrusion alerts.

AetherShield VMS	2026
AI / Computer Vision Hackathon Project	AI Fence GitHub Live

- AI video-management and security platform for real-time object detection, tracking, identity classification, and zone-intrusion monitoring; deployed YOLOv11n + ByteTrack at 0.45 confidence / 200 ms, tracking 11 COCO classes across rectangle, circle, and polygon ROIs.
- Integrated Haar-cascade face matching with 64×64 histogram embeddings and cosine similarity threshold 0.72 for known/unknown/blacklist identity classification on person tracks.
- Enabled natural-language video search and video QA using structured detection metadata including class, track ID, zone, and timestamp instead of raw video frames.

Projects

ProductForge	GitHub Live
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- AI product-management platform that converts startup ideas into market analysis, PRDs, user stories, and prioritized roadmaps; used Claude Sonnet 4 / GPT-4o with Generative AI, LLM prompting, and structured outputs for TAM/SAM/SOM, competitors, revenue model, risks, and GTM.
- Generated PRDs, 8–12 Given/When/Then user stories, and Now/Next/Later roadmaps; prioritized features using RICE scoring for Reach/Impact, Confidence, and Effort with MoSCoW prioritization.

SCHOLAR	GitHub Live
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- AI research assistant for querying academic papers, retrieving evidence, and identifying research gaps; extracted papers into schema-locked JSON and indexed 1,200-character chunks with 200 overlap using 384-dim BAAI/bge-small-en-v1.5 embeddings.
- Implemented **RAG, vector search, cosine similarity, and retrieval-augmented question answering** using top-6 hits with an explicit “not in this paper” refusal; added multi-paper gap analysis, hypothesis generation, and literature synthesis.

PathWise	GitHub Live
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- AI career intelligence platform for resume screening, job matching, ATS optimization, and interview feedback; designed a weighted ATS scoring model using JD keywords (32%), sections (20%), skills (16%), and quantified bullets (10)
- Matched resumes to roles using an O*NET-style skill corpus; used Gemini/OpenAI/Anthropic for copy rewriting and evaluated interviews using technical signal, STAR structure, length, and communication scoring.